Ben Clingenpeel

Washington, DC
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Education

2023-present | **George Washington University**
PhD in Mathematics

2019-2023 | **Georgetown University**
Bachelor of Science in Mathematics

Interests

I am a third year PhD student at George Washington University and am interested in pursuing a career in mathematical research and teaching at the college level. I am interested in knot theory and my advisor is Professor Józef H. Przytycki.

Preprints

2025 | **In search of homology for quasigroups of Bol-Moufang type.** With Anthony Christiana, Huizheng (Ali) Guo, Jinseok Oh, Józef H. Przytycki, and Anna Zamojska-Dzieno. arXiv:2508.21268.

Publications

2026 | **Colorings of symmetric unions and partial knots.** With Zongzheng Dai, Gabriel Diraviam, Kareem Jaber, Krishnendu Kar, Ziyun Liu, Teo Miklethun, Haritha Nagampoozhy, Michael Perry, Moses Samuelson-Lynn, Eli Seamans, Ana Wright, Nicole Xie, Ruiqi Zou, and Alexander Zupan. arXiv:2504.08589.
Journal of Knot Theory and Its Ramifications, Vol 35, No. 06.

2025 | **Low dimensional homology of the Yang-Baxter operators yielding the HOMFLYPT polynomial.** With Anthony Christiana, Huizheng (Ali) Guo, Jinseok Oh, Józef H. Przytycki, Xiao Wang, and Hongdae Yun. arXiv:2502.20659.
To appear in *Homology, Homotopy, and Applications*.

Presentations

April 2026	**Symmetrically related knots.** Poster presented at the George Washington University Columbian College of Arts & Sciences annual Research Showcase.
November 2025	**Symmetric unions of knots.** Talk given at the George Washington University Graduate Student Seminar.
October 2025	**Computing homology of the Yang-Baxter operators yielding the HOMFLYPT polynomial.** Given at the Special Session on Inverse Problems of the AMS Fall Central Sectional Meeting in St. Louis, MO.
October 2025	**Homology of Yang-Baxter Operators.** Specialty exam talk given at George Washington University.
May 2025	**Low dimensional homology of the Yang-Baxter operators yielding the HOMFLYPT polynomial.** Given at the Special Session on Yang-Baxter operators in Knot Theory of the AMS Spring Western Sectional Meeting in San Luis Obispo, CA.
April 2025	**Low dimensional homology of the Yang-Baxter operators yielding the HOMFLYPT polynomial.** Distinguished Graduate Student talk given at *Knots in Washington 51* at the George Washington University in Washington, DC.
January 2023	**Colorings of knots with symmetric union presentations.** Talk given with Eli Seamans at the special session of the Joint Mathematics Meeting on the Polymath Jr program in Boston, MA.

Teaching and Tutoring

July 2024 - Present **Instructor of Record** | George Washington University

- Taught MATH 1231, Calculus I (Summer 2024).
- Taught MATH 1232, Calculus II (Summer 2025).

Aug 2023 - Present **Graduate Teaching Assistant** | George Washington University

- Graded and taught recitation sections for MATH 1231, Calculus I (Fall 2023, Spring 2024, Fall 2024).
- Graded and taught recitation sections for MATH 1232, Calculus II (Spring 2025).
- Graded and taught recitation sections for MATH 2184, Linear Algbera I (Fall 2025, Spring 2026).
- Tutored students in lower division math courses through the Calculus Lab (all semesters).

Jan 2022 - May 2023 **Math Tutor** | Georgetown University Math Assistance Center

- Tutored Georgetown University students in lower division courses.
- Led trainings for new tutors.

Aug 2021 - Dec 2022 **Math Teaching Assistant and Grader** | Georgetown University

- Graded student assignments in MATH 215, Abstract Algebra (Spring 2022, Fall 2022).
- Graded student assignments in MATH 150, Linear Algebra and held a weekly office hour (Fall 2021).

Jan 2020 - May 2020 **Math Tutor** | Georgetown University GUMSHOE Program

- Volunteered with a campus math and science tutoring group.
- Worked with middle school students from DCs seventh and eighth wards who came to campus for extra weekend math lessons.

Sep 2017 - Present **Math Tutor** | Freelance

- Tutored students at a variety of different levels, some in middle school and high school classes, as well as a few in first college calculus courses.
- Designed curricula for students who weren't following a given course schedule. For these students, I selected appropriate topics in math that don't typically appear in middle school and high school curricula.

Conference Activities

Name	**Location**	**Role(s)**
Knots in Washington 53 (April 24-26, 2026)	Washington, DC (George Washington University)	Co-organizer
Knots in Washington 52 (December 5-7, 2025)	Washington, DC (George Washington University)	Co-organizer
AMS 2025 Fall Central Sectional Meeting	St. Louis, MO (Saint Louis University)	Speaker: *Computing homology of Yang-Baxter operators yielding the HOMFLYPT polynomial*
AMS 2025 Spring Western Sectional Meeting	San Luis Obispo, CA (California Polytechnic State University)	Speaker: *Low dimensional homology of Yang-Baxter operators yielding the HOMFLYPT polynomial*

Knots in Washington 51 (April 4-6, 2025)	Washington, DC (George Washington University)	Distinguished Graduate Student Speaker: *Low dimensional homology of Yang-Baxter operators yielding the HOMFLYPT polynomial* Co-organizer
Knots in Washington 50 (December 6-8, 2025)	Washington, DC (George Washington University)	Co-organizer
Knots in Washington 49.984375 (April 26-28, 2024)	Washington, DC (George Washington University)	Co-organizer
AMS 2024 Spring Eastern Sectional Meeting	Washington, DC (Howard University)	Attendee
Knots in Washington 49.96875 (December 8-10, 2023)	Washington, DC (George Washington University)	Co-organizer
Joint Mathematics Meetings 2023	Boston, MA	Speaker: *Colorings of knots with symmetric union presentations*

Honors and Awards

2026 **Marvin Green Prize.** Awarded to an outstanding graduate or undergraduate student who has made significant use of computing in his or her work. *George Washington University.*

2025 **Distinguished Graduate Student Speaker.** *Knots in Washington 51.*

2023 **Henry M. Leslie Award.** Awarded to that member of the senior class showing highest proficiency in Mathematics. *Georgetown University.*